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Chapter 8

Biomolecules

5SLXJ5

Learning Objectives

The learner will be able to,

- *List out the inorganic and organic components of a cell.*
- *Understand about bonding pattern of water and properties of water.*
- *Familiarise with the classification of carbohydrates and its functions.*
- *Recognise the basic structure of carbohydrates, proteins, lipids and nucleic acids and differentiate the various pattern of classification with respect to structure.*
- *Identify the structure and functions of carbohydrates.*
- *Familiarise with the general structure of amino acids and its classification based on the functional group.*
- *Comparative study of the primary, secondary, tertiary and quaternary structure of proteins.*
- *Know the structure and classification of enzymes.*
- *Know about the factors affecting the mode of action of enzymes with relevant examples.*
- *Understand lipids as a biomolecule and discuss the properties of lipids.*
- *Have a deeper knowledge about structure of nucleic acids.*
- *Recognize nucleic acids as a polymer which plays a vital role in carrying the genetic information.*
- *Learn about the different forms of DNA and types of RNA.*

Chapter Outline

8.1 Water

8.2 Primary and Secondary Metabolites

8.3 Carbohydrates – Classification and Structure

8.4 Lipids – Classification and Structure

8.5 Proteins and Amino Acids – Classification and Structure,

8.6 Enzymes – Classification, Nomenclature, Structure and Concepts, Mechanism of Enzyme Action, Activation energy, factors affecting enzyme action.

8.7 Nucleic Acids general Structure and composition – Forms of DNA and Types of RNA.